Error-Driven Context Resets Are a Tail Risk for Tabular In-Context Learners on Data Streams

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Abstract

Tabular foundation models (TFMs) such as TabPFN, TabICL and TabDPT predict from a context of labelled rows without gradient updates, which makes them attractive for data streams: keeping the most recent rows in the context (FIFO) is already a form of adaptation. A long line of stream-learning work adds an error-driven drift detector that, on an alarm, discards the past. We ask whether this practice transfers to in-context learners. On 39 real streams from 19 independent sources and 42 synthetic streams, with eight detectors, three reset actions and three TFMs, we find that a full context reset is a *tail risk*: averaged over detectors it loses 26–31 accuracy points on the worst source while gaining only about 5–6 on the best. Most single resets that change the outcome on real streams are net losses: 87–88% of all such resets, and the majority on 15 of the 19 sources. Over sources, no detector rejects the signed-rank null after Holm correction, so the practice is neither uniformly harmful nor useful; its risk sits in the tail. Two observations are consistent with the asymmetry. Under a FIFO budget, a reset changes the context of at most the next $\lceil M/B \rceil - 1$ batches (we prove this and use it for an exact per-reset allocation of the difference to FIFO), so the window already forgets and leaves a reset little to gain; trained incremental learners, which never forget, gain from the same resets (+1.8 and +7.3 points on average). We then show that hedging the reset context against the FIFO context with exponential weights substantially reduces the tail at little cost: we prove a cumulative log-loss guarantee of at most $\ln 2/\eta$ per detection relative to FIFO for the undiscounted variant with $0 < \eta \leq 1$, and empirically the worst source improves to about -2 points while 90% of the best-source gain and the synthetic gains are kept; on one of the two batches of added sources the hedge still loses 0.4–0.9 points on average. The findings hold across backbones, backbone seeds, context budgets and hedge hyperparameters.

1 Introduction

Tabular foundation models are transformers pre-trained on synthetic tasks that solve a new classification problem in a single forward pass over a *context* of labelled rows (Müller et al., 2022; Hollmann et al., 2023; 2025; Qu et al., 2025; Ma et al., 2024). On a data stream, the natural way to use such a model is to keep the most recent M labelled rows as its context and to predict each incoming batch from them. The model is frozen, so the only thing that adapts is the context, and a first-in-first-out (FIFO) window already forgets old concepts at a fixed rate.

Classical stream learning adapts in a different way. A learner is trained incrementally, and a drift detector monitors its error rate; when the detector fires, the learner is reset and retrained on recent data (Gama et al., 2004; Baena-García et al., 2006; Bifet & Gavaldà, 2007; Frías-Blanco et al., 2015; Pesaranghader & Viktor, 2016; Raab et al., 2020). This detect-and-reset pattern is the default in surveys and libraries (Gama et al., 2014; Lu et al., 2019; Montiel et al., 2021). For an in-context learner its analogue is simple: on an alarm, empty the context and refill it from the latest labelled batch. It is tempting to bolt this onto a TFM, and doing so costs nothing to implement. We ask whether it should be done.

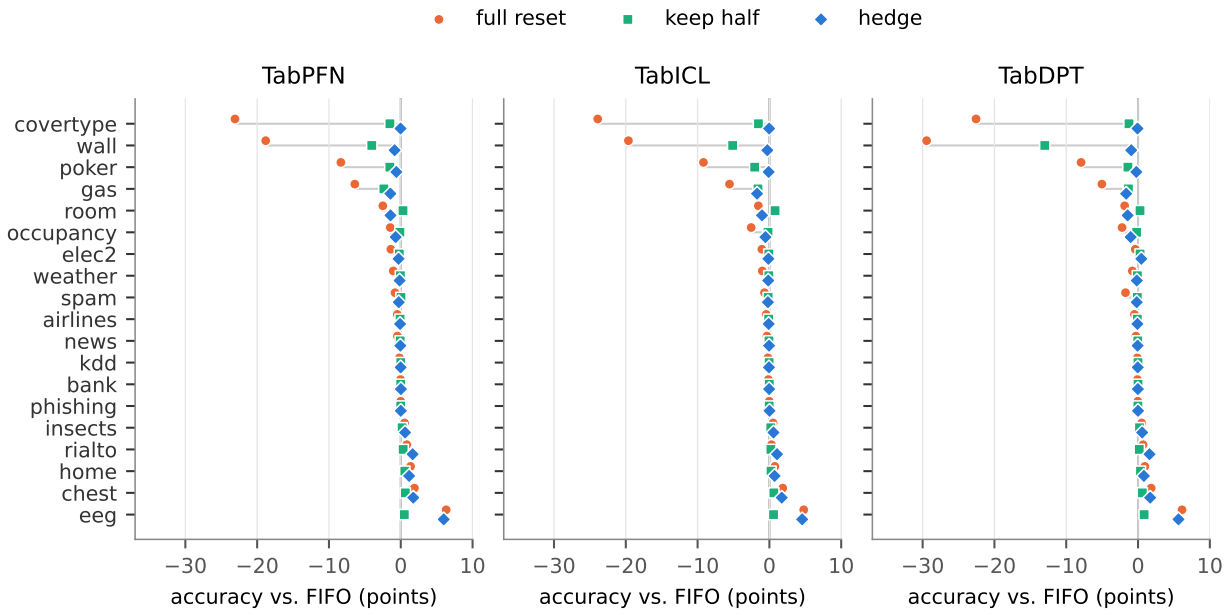


Figure 1: Effect of the three actions on detection, per real source and averaged over the eight detectors, as accuracy minus FIFO with the same TFM ($M = 1000$). A full reset (orange) loses heavily on a few sources (covertype, wall, poker, gas) and gains on a few others (eeg, chest, home); hedging against FIFO (blue) reduces the losses to at most a few points and keeps most of the gains. Sources are ordered by the full-reset effect on TabPFN. The figure averages over detectors first (covertype: −23.1 for TabPFN); the worst-source statistic of the text and Table 2 takes each detector's worst source and then averages (−26.2), so the two need not coincide.

We run the experiment at scale (Section 4): 81 streams (39 real ones from 19 independent sources and 42 synthetic ones), eight error-driven detectors with their default parameters, three actions on detection (full reset, keep half of the window, and a hedge described below), three TFMs (TabPFN v2, TabICL, TabDPT), three backbone seeds, three context budgets, and two trained incremental learners for contrast. Every comparison is against the same model with a FIFO window, and every test treats the 19 real sources, not the 39 streams, as the independent units. Three observations stand out (Figure 1).

Observation 1 (Resets carry a heavy left tail)**.** Averaged over the eight detectors, a full reset loses 26.2, 26.5 and 31.1 accuracy points against FIFO on the worst real source for TabPFN, TabICL and TabDPT, and gains at most 6.3, 4.8 and 6.1 points on the best. Of the single resets that change the accuracy of the batches they affect, 87–88% are net losses on real streams (pooled over all resets; 69–73% when each source is weighted equally, and more than half on 15 of 19 sources).

Observation 2 (The average hides the tail)**.** Over the 19 real sources the mean effect of a full reset is negative for every detector and backbone, but no detector rejects the signed-rank null after Holm correction, and resets help on several sources (eeg, chest, home). On synthetic streams with abrupt, recurring concepts resets help by about 8 points.

Observation 3 (A FIFO window leaves a reset little to gain)**.** The same detectors on trained learners that never forget gain on average (Hoeffding tree +1.8, Gaussian naive Bayes +7.3 points), where TabPFN loses (−2.8). On the sources where resets help, they help the trained learners several times more (eeg: +28.5 and +39.2 against +6.3). For TabPFN the best-source gain grows with the context budget (4.4, 6.3 and 7.7 points for $M = 500, 1000, 2000$).

These observations suggest a reading of the question. A detector is useful when it fires on a real change and costly when it fires on noise; a FIFO window plausibly captures much of the benefit of the first case, while a reset pays the full cost of the second. We do not observe which alarms on real streams are correct, so this

is an interpretation consistent with the results rather than an identified mechanism. The practical question is therefore not whether drift detection is good or bad for TFMs, but

> *How can an in-context learner on a stream act on a drift alarm without exposing itself to the tail risk of a false one?*

Remark 1. We do not propose a new detector or a new stream learner, and we do not claim state-of-the-art accuracy. Our objective is to establish, with controlled comparisons and source-level statistics, what error-driven resets do to tabular in-context learners, what plausibly explains it, and which minimal change makes them much safer to use.

Our answer is to keep the FIFO context alive next to the reset one and to mix their predictions with exponential weights on their recent log-loss (Vovk, 1990; Cesa-Bianchi & Lugosi, 2006). For the undiscounted variant a false alarm then costs at most a constant in cumulative log-loss, while the weights move to the reset context when it predicts better.

Contributions.

1. **A protocol with exact accounting.** We show that under a FIFO budget a reset changes the context of at most $\lceil M/B \rceil - 1$ batches (Proposition 1), so the realised difference to FIFO decomposes exactly into per-reset contributions.
2. **A large controlled evaluation.** Eight detectors × three actions × three TFMs on 81 streams, with backbone seeds, context budgets and two trained learners, and tests clustered by source (Section 5). The supplementary material holds the code, the stream construction scripts and the per-run results, from which every table except the list of sources, the figure and the check of Theorem 1(ii) are regenerated; the cached predictions will be released with the paper.
3. **A finding and a plausible explanation.** Error-driven resets are a tail risk for TFMs, not a uniform loss (Observations 1 and 2); the evidence is consistent with a FIFO window absorbing much of what a reset could gain (Observation 3, Proposition 1).
4. **A minimal remedy with a guarantee.** For $\eta \leq 1$ without discounting, hedging the reset context against FIFO is never worse than FIFO by more than $\ln 2/\eta$ per detection in cumulative log-loss (Theorem 1); empirically it reduces the worst-source loss from 26–31 to about 2 points on all three backbones while keeping most of the gains, and the tested TabPFN configurations of its two hyperparameters behave alike.

2 Setting

Stream protocol. A stream is a sequence of labelled rows $(x_1, y_1), (x_2, y_2), \ldots$ with $y_i \in \{1, \ldots, K\}$, cut into batches of B rows; batch t contains rows $tB + 1, \ldots, (t + 1)B$. Batch t is predicted after the labels of all earlier batches have arrived and before its own labels arrive (labels are one batch late). Batch 0 is not scored. We use $B = 100$.

Definition 1 (Context policy)**.** A *context policy* chooses, before batch t is predicted, a start index $\mathrm{lo}_t \leq tB$; the context is the contiguous window of labelled rows

$$\mathcal{C}_t = \{(x_i, y_i) : \mathrm{lo}_t < i \leq tB\}, \qquad tB - \mathrm{lo}_t \leq M, \tag{1}$$

where M is the context budget. The FIFO policy uses $\mathrm{lo}_t^F = \max(0, tB - M)$.

A TFM f maps a context and a query row to a predictive distribution $p_t(\cdot \mid x) = f(\mathcal{C}_t, x)$.

Assumption 1 (Frozen in-context learner)**.** The model is not updated on the stream, and for a fixed backbone seed its prediction is a deterministic function of the context and the query row.

Assumption 1 holds for the three TFMs we use (we verified it by repeated prediction). It has a useful consequence: since every policy in this paper uses a contiguous window, every prediction is identified by the pair (lo_t, t), and predictions can be cached and shared between policies without changing any of them.

Error-driven resets. A detector receives the 0/1 errors of the policy's own predictions, row by row, after the labels of a batch arrive. If it fires while processing the errors of batch $t-1$, the rest of that batch is not fed and one of three actions is taken before batch t is predicted:

- RESET: the context restarts at the latest labelled batch, $s \leftarrow (t-1)B$, and $\mathrm{lo}_u = \max(s, \mathrm{lo}_u^F)$ for $u \geq t$ until the next detection;
- HALF: the context keeps the latest $M/2$ rows, $s \leftarrow tB - M/2$;
- HEDGE: both the FIFO context and the RESET context are kept, and the prediction mixes them (Section 3.2).

The eight detectors are DDM (Gama et al., 2004), EDDM (Baena-García et al., 2006), FHDDM (Pesaranghader & Viktor, 2016), HDDM_A and HDDM_W (Frías-Blanco et al., 2015), ADWIN (Bifet & Gavaldà, 2007), Page–Hinkley (Page, 1954) and KSWIN (Raab et al., 2020), all in their `river` 0.26.1 implementation with default parameters (Montiel et al., 2021). No parameter of any detector or action was tuned.

3 Why a FIFO window limits what a reset can do

This section has two steps. Proposition 1 shows that a reset is a temporary perturbation of the FIFO context of a known, bounded length; this gives an exact accounting of every reset and caps what a single reset can gain or lose. Theorem 1 then shows that mixing the reset context with the FIFO context bounds the cost of any reset in cumulative log-loss by a constant.

3.1 The finite memory of a reset

Let $H = \lceil M/B \rceil$ be the number of batches the FIFO window spans. For a full reset at batch τ write $s_\tau = (\tau - 1)B$.

Proposition 1 (Finite memory of a reset)**.** *Under Assumption 1, consider the* RESET *policy with detections at batches $\tau_1 < \tau_2 < \cdots < \tau_N$ (τ_{N+1} is the end of the stream). Then:*

(i) for $\tau_j \leq t < \tau_{j+1}$ the context differs from the FIFO context only if $t \leq \tau_j + H - 2$, i.e. a reset changes the context of at most $H-1$ batches;

(ii) with a_t and a_t^F the accuracies of batch t under RESET *and FIFO, the total difference decomposes exactly into per-reset effects,*

$$\sum_{t \geq 1} \left(a_t - a_t^F\right) = \sum_{j=1}^{N} \Delta_j, \qquad \Delta_j = \sum_{t=\tau_j}^{\min(\tau_j + H - 1,\, \tau_{j+1}) - 1} \left(a_t - a_t^F\right); \tag{2}$$

(iii) each per-reset effect is bounded, $|\Delta_j| \leq H - 1$ (in batches of accuracy).

Remark 2. Part (ii) makes the per-reset statistics of Section 5 an exact allocation of the realised difference to FIFO rather than an estimate: each Δ_j is read off from the FIFO run on the same batches. It is not the counterfactual effect of removing alarm j alone, which could change later errors, detector states and alarms. Part (iii) offers an explanation for Observation 3; the bound is symmetric in gains and losses, so it does not by itself explain their asymmetry. A reset can only be ahead of FIFO for $H-1$ batches; after that the window has forgotten the old concept on its own. A trained learner that never forgets has no such cap, so the same correct alarm is worth more to it. The cap grows with M, which matches the growth of the best-source gain with the budget. A false alarm also costs at most $H-1$ batches, but the costs of frequent false alarms add up, while the gain of a correct alarm is capped.

3.2 Hedging the reset against FIFO

The HEDGE action keeps two experts: F, the FIFO context, and R, the context of the RESET policy driven by the same detections. For a batch t and an expert $k \in \{F, R\}$ write $p_{k,t,i}$ for the probability it assigns to the true label of row i of batch t, and

$$\ell_{k,t} = -\frac{1}{B} \sum_{i=1}^{B} \log p_{k,t,i} \tag{3}$$

for its mean log-loss on the batch. The detections split the stream into segments $S_0, S_1, \ldots, S_N$, where S_j runs from batch τ_j up to batch $\tau_{j+1} - 1$ ($\tau_0 = 1$). Within a segment the weights are

$$w_{k,t} \propto \exp\Big(-\eta \sum_{u \in S_j,\, u<t} \gamma^{\,t-1-u}\, \ell_{k,u}\Big), \qquad p_{\mathrm{mix},t,i} = w_{F,t}\, p_{F,t,i} + w_{R,t}\, p_{R,t,i}, \tag{4}$$

so a detection restarts R and resets both weights to $1/2$. Before the first detection the two experts coincide and the mixture is FIFO. We use $\eta = 2$ and $\gamma = 0.5$, the values of a sliding-window ensemble rule we used before this study; Section 5.4 shows that the results do not depend on them.

Assumption 2 (Positive probabilities)**.** Every expert assigns positive probability to the true label of every row, $p_{k,t,i} > 0$.

Theorem 1 (Hedging bounds the cost of a reset)**.** *Under Assumptions 1 and 2, let $\eta \in (0, 1]$ and $\gamma = 1$. Then:*

(i) on every segment the mixture is close to the better expert of that segment,

$$\sum_{t \in S_j} \ell_{\mathrm{mix},t} \;\le\; \min\Big\{ \sum_{t \in S_j} \ell_{F,t},\; \sum_{t \in S_j} \ell_{R,t} \Big\} + \frac{\ln 2}{\eta}; \tag{5}$$

(ii) over the whole stream with N detections,

$$\sum_{t \ge 1} \ell_{\mathrm{mix},t} \;\le\; \sum_{t \ge 1} \ell_{F,t} + \frac{N \ln 2}{\eta} \quad \textit{and} \quad \sum_{t \ge 1} \ell_{\mathrm{mix},t} \;\le\; \sum_{j=0}^{N} \min_{k \in \{F,R\}} \sum_{t \in S_j} \ell_{k,t} + \frac{N \ln 2}{\eta}. \tag{6}$$

Remark 3. The first bound in (ii) is the safety guarantee: whatever the detector does, the hedge loses at most a constant per alarm relative to never resetting in cumulative log-loss. Here ℓ is the mean log-loss of a batch, so in log-loss summed over rows the allowance is $BN \ln 2/\eta$. This bound grows with the number of alarms and does not guarantee a small accuracy loss; the accuracy-tail reduction is an empirical finding. The second bound is the benefit: on segments where the reset context is better, the hedge follows it up to the same constant. The proof (Appendix A) is the aggregating-algorithm argument (Vovk, 1990) with one change: the weights are updated once per batch rather than once per row, and a generalised Hölder inequality shows that this costs nothing. The bound is stated for $\eta \le 1$ and no discounting; with $\gamma < 1$ the weights forget faster and our argument does not give a bound of this form, which is why we also report the variant $\eta = 1, \gamma = 1$. The implementation clips probabilities at 10^{-6} for loss updates and normalizes the cached mixture before scoring. The theorem concerns positive, normalized expert probabilities without this clipping; the numerical check below is a check of the implemented variant on these streams, not a proof that clipping preserves the bound on arbitrary streams.

4 Experimental setup

Streams. The 39 real streams come from 19 sources (Appendix B): covertype, poker, airlines (each also as several contiguous segments), the INSECTS streams (Souza et al., 2020), electricity (Harries, 1999), phishing, rialto, spam, weather, and ten further public sources added after the first analyses in two batches of five (gas, occupancy, room, bank, kdd; then eeg, news, home, wall, chest). For each batch the selection rule and preprocessing were fixed before download, but the second batch was added after seeing the first, so

Table 1: Effect of error-driven resets on TabPFN ($M = 1000$): accuracy of each policy minus FIFO, in points. Real: source-weighted mean over 19 sources with 95% cluster-bootstrap interval, and Holm-corrected Wilcoxon p over sources for the full reset. Synthetic: mean over 7 generator families.

	Real (19 sources)				Synthetic (7 families)		
Detector	RESET	p_{Holm}	HALF	HEDGE	RESET	HALF	HEDGE
DDM	−0.82 [−2.3, +0.6]	0.62	−0.36	+0.16 [−0.5, +1.1]	+9.40	+1.82	+9.17
EDDM	−3.71 [−7.8, −0.4]	0.36	−0.48	+0.34 [−0.3, +1.2]	+8.96	+3.60	+9.04
FHDDM	−2.33 [−5.2, −0.3]	0.37	−0.13	+0.13 [−0.3, +0.7]	+0.93	+0.51	+1.16
HDDM_A	−3.93 [−8.8, −0.1]	0.62	−0.82	+0.20 [−0.5, +1.1]	+9.43	+1.68	+9.16
HDDM_W	−3.33 [−7.3, −0.1]	0.56	−0.32	+0.30 [−0.3, +1.1]	+9.21	+1.65	+8.96
ADWIN	−3.21 [−7.5, +0.1]	0.62	−0.48	+0.30 [−0.3, +1.1]	+9.10	+1.97	+8.97
Page–Hinkley	−2.41 [−5.9, +0.2]	0.62	−0.21	+0.46 [−0.1, +1.2]	+8.28	+2.23	+8.30
KSWIN	−3.00 [−6.7, 0.0]	0.62	−0.31	+0.29 [−0.3, +1.1]	+8.83	+1.68	+8.52

tests over all 19 sources are sequential and exploratory; we report the batches separately as well (Section 5 and Appendix F). Segments of one data set are not independent, so all statistics on real streams are computed over the 19 sources: a source's value is the mean over its streams. The 42 synthetic streams are the test seeds of six `river` generators (Agrawal, hyperplane, RBF, SEA, sine, STAGGER; five seeds each) and twelve grid streams with three abrupt, recurring block concepts (RBF mixtures with 5, 30 or 100 centroids whose labels are permuted between concepts, blocks of 500 or 2000 rows, two seeds); synthetic values are averaged per generator family.

Models. TabPFN v2 (Hollmann et al., 2025), TabICL (Qu et al., 2025) and TabDPT (Ma et al., 2024), each with four ensemble members, $M = 1000$ unless stated otherwise. As trained learners we use the `river` Hoeffding tree (Domingos & Hulten, 2000) and Gaussian naive Bayes with the same protocol; their reset replaces the model by a fresh one that learns from the latest labelled batch on, and their hedge mixes the never-reset and the reset model with the same rule.

Measures. For a policy and a stream, d is its accuracy minus the accuracy of FIFO with the same model, in points. We report source-weighted means over the 19 real sources with 95% cluster-bootstrap percentile intervals (10 000 resamples of sources, pointwise and not adjusted for multiplicity) and two-sided Wilcoxon signed-rank tests over source means (zero differences dropped) with Holm correction over the eight detectors of each action (Holm, 1979). The signed-rank test concerns the median-type location of the source effects, not their mean. Because the risk of resets is in the tail, we also report the *worst* and *best* source (for each detector the minimum and maximum over sources, averaged over detectors). Single-reset effects are Δ_j of Proposition 1. Beyond accuracy we report macro-F1, macro one-vs-rest ROC AUC, expected calibration error (ECE, 15 bins), log-loss and Brier score, recovered by replaying each policy from the cached predictions (Appendix D).

Pre-registration. Every stage was designed before it was run, with written hypotheses and pass criteria; Appendix F lists them with their outcomes, including the ones that failed.

5 Results

5.1 Full resets: a heavy left tail, no reliable average

Table 1 gives the source-weighted effect of every detector and action for TabPFN. On real streams a full reset is worse than FIFO on average for all eight detectors, between −0.8 and −3.9 points, but the bootstrap intervals are wide, five or six of the 19 sources benefit, and after Holm correction no detector rejects the signed-rank null ($p_{\text{Holm}} \geq 0.36$). The two batches of added sources differ: on the first (gas, occupancy, room, bank, kdd) a full reset loses on four or five of the five sources for every detector (−1.8 to −2.5 points), on the second (eeg, news, home, wall, chest) it helps on three of the five (detector means from −4.7 to +1.6).

Table 2: The tail of resets on three TFMs ($M = 1000$), accuracy minus FIFO in points. Worst/best: for each detector the worst/best of the 19 real sources, averaged over the 8 detectors. Harm: share (%) of the single resets with a non-zero effect Δ_j (Proposition 1) that are net losses, on real streams, pooled over all resets of all streams and detectors (source-weighted shares are in the text).

	RESET		HALF		HEDGE		Real mean			Synthetic	
Model	worst	best	worst	best	worst	best	RESET	HEDGE	Harm	RESET	HEDGE
TabPFN	−26.2	+6.3	−4.3	+0.6	−1.6	+6.0	−2.8	+0.3	88.1	+8.0	+7.9
TabICL	−26.5	+4.8	−5.4	+0.9	−1.7	+4.6	−3.1	+0.2	88.4	+8.4	+8.3
TabDPT	−31.1	+6.1	−13.0	+0.9	−2.1	+5.6	−3.3	+0.3	87.3	+8.2	+8.1

Table 3: The same detectors on trained incremental learners and on TabPFN (all 81 streams), accuracy minus the never-reset learner in points, averaged over the 8 detectors. For trained learners, Harm measures negative non-zero differences within the first ten batches after an alarm (cut at the next alarm), not the full effect of the reset. For TabPFN, Harm uses the exact finite-memory attribution in Table 2.

	RESET					HEDGE			
Learner	real	worst	best	synth.	Harm	real	worst	best	synth.
Naive Bayes	+7.3	−11.7	+40.1	+13.7	48.8	+8.3	−6.9	+35.6	+12.8
Hoeffding tree	+1.8	−17.5	+28.5	+12.5	69.9	+3.5	−7.8	+25.3	+13.7
TabPFN	−2.8	−26.2	+6.3	+8.0	88.1	+0.3	−1.6	+6.0	+7.9

On synthetic streams the picture is reversed: full resets gain up to 9.4 points (8.0 averaged over detectors; FHDDM, which fires least, gains 0.9), because the concepts change abruptly and recur, which is the situation detectors were designed for.

The average hides where the risk is. Table 2 reports, for each backbone, the worst and the best source of each action. A full reset loses 26 to 31 points on the worst source and gains approximately 5 to 6 on the best. The loss is not a property of one model: covertype and wall are the two worst sources for all three TFMs. At the level of single resets (Proposition 1), 88.1% (TabPFN), 88.4% (TabICL) and 87.3% (TabDPT) of the resets that change the accuracy of the batches they affect are net losses on real streams, against 11.5–15.4% on synthetic streams. These shares pool all resets of all streams and detectors, so sources with many alarms weigh more. Weighting each source equally gives 69.5%, 72.9% and 68.8%; the share exceeds one half on 15 of the 19 sources for every backbone, and it lies between 81.6% and 90.7% for every detector and backbone. For TabPFN, 25 699 resets lose, 3 460 gain and 1 057 leave the accuracy unchanged.

5.2 A FIFO window already forgets: evidence from trained learners

Table 3 applies the same detectors to two trained incremental learners. For them resets help on average (+1.8 for the Hoeffding tree, +7.3 for naive Bayes). Within the first ten batches after each alarm, cut at the next alarm, 70% and 49% of their non-zero accuracy differences are negative. These local-window statistics are descriptive: unlike TabPFN's exact finite-memory attribution (88% harmful), they do not capture the full effects of trained-model resets, which can persist longer. The losses on the worst sources are of the same order for all learners (covertype and wall again); what differs is the gain. On eeg, chest and home, where resets help most, they gain 28.5, 14.5 and 12.5 points for the Hoeffding tree, 39.2, 23.4 and 34.1 for naive Bayes, and only 6.3, 1.9 and 1.4 for TabPFN. A learner that never forgets needs the reset to forget; a FIFO window does it anyway, and Proposition 1(iii) caps the advantage a reset can build. The weaker and the less forgetful the learner, the more resets help (naive Bayes > Hoeffding tree > TabPFN). The trained learners are also far less accurate in absolute terms (72.4% and 61.9% against 87.8% for TabPFN on real sources), so part of their gain reflects a lower starting point (Appendix E).

The context budget moves the cap. With $M = 500$ the best-source gain of a full reset shrinks to 4.4 points and its worst-source loss to −17.6; with $M = 2000$ the gain grows to 7.7 while the loss stays at −26.6

Table 4: TabPFN with three context budgets, accuracy minus FIFO in points (detector average; real: 19 sources). $H-1$ is the number of batches a reset can affect (Proposition 1).

		RESET					HEDGE		
M	$H-1$	real	worst	best	Harm	synth.	real	worst	best
500	4	−2.0	−17.6	+4.4	80.5	+3.5	0.0	−2.6	+3.4
1000	9	−2.8	−26.2	+6.3	88.1	+8.0	+0.3	−1.6	+6.0
2000	19	−2.8	−26.6	+7.7	88.1	+14.6	+0.5	−1.9	+7.1

Table 5: Metrics beyond accuracy on the 19 real sources ($M = 1000$; source-weighted means; reset actions averaged over the 8 detectors). Accuracy, macro-F1, macro one-vs-rest ROC AUC and ECE in %.

Model	Policy	Acc.	F1	AUC	ECE	Log-loss	Brier
TabPFN	FIFO	87.76	79.89	90.60	2.39	0.392	0.175
	RESET	84.91	76.39	87.82	3.57	0.617	0.217
	HALF	87.37	79.50	90.31	2.55	0.415	0.181
	HEDGE	88.03	80.39	90.74	2.16	0.380	0.171
TabICL	FIFO	88.05	80.43	90.79	2.99	0.387	0.173
	RESET	85.00	76.64	87.90	4.14	0.613	0.217
	HALF	87.61	79.92	90.39	3.12	0.415	0.179
	HEDGE	88.27	80.91	90.85	2.67	0.376	0.169
TabDPT	FIFO	87.41	79.57	90.49	2.70	0.399	0.181
	RESET	84.10	75.56	87.17	4.16	0.647	0.229
	HALF	86.63	78.56	89.75	2.87	0.441	0.192
	HEDGE	87.66	80.02	90.61	2.25	0.387	0.176

(Table 4). On synthetic streams, where most non-zero reset effects are beneficial, the gain grows from 3.5 to 8.0 to 14.6 points. The gain side follows Proposition 1(iii): a longer window forgets more slowly, so a correct reset stays ahead for longer. The loss side does not grow beyond $M = 1000$, plausibly because on the worst sources a reset at $M = 1000$ already discards nearly all useful context. The hedge keeps the synthetic gain at every budget (+14.3 at $M = 2000$).

5.3 The hedge reduces the observed tail

Keeping FIFO next to the reset context changes the picture on every backbone (Table 2): the worst source improves from −26 to −31 points to about −2, the best source keeps 90% of the gain of a full reset or more (e.g. +6.0 vs. +6.3 for TabPFN), and on synthetic streams the hedge keeps essentially all of the gain of a full reset (+7.9 vs. +8.0). Its average on real streams is small and positive (+0.2 to +0.3 points on the three backbones), not significantly different from FIFO; its value is the substantial reduction of the observed worst-source loss. It is not free everywhere: on the first batch of added sources (gas, occupancy, room, bank, kdd) its source-weighted mean is −0.4 to −0.9 points, below the pre-registered −0.2 (Appendix F). Keeping half of the window (HALF) is a weaker compromise: it shortens the tail to −4.3, −5.4 and −13.0 points but gives up most of the gain (best source below +1). The hedge helps the trained learners as well (Table 3), so it is a general way to act on an alarm, not a fix specific to TFMs.

Table 5 shows that the same ordering holds for macro-F1, ROC AUC, calibration and log-loss on all three backbones. Full resets are worse on every measure (for TabPFN, macro-F1 76.4 against 79.9, ECE 3.6% against 2.4%, log-loss 0.62 against 0.39), and the hedge is slightly better than FIFO on every measure (macro-F1 80.4, ECE 2.2%, log-loss 0.38).

Table 6: Sensitivity of the hedge to η and γ (TabPFN, $M = 1000$; detector average, points). For reference, the full reset has real mean -2.84, worst source -26.23, best source $+6.32$ and synthetic mean $+8.02$.

η	γ	real	worst	best	synth.
2	0.5 (default)	$+0.27$	-1.64	$+5.98$	$+7.91$
0.5	0.5	$+0.24$	-1.63	$+5.89$	$+7.86$
1	0.5	$+0.25$	-1.62	$+5.95$	$+7.90$
4	0.5	$+0.27$	-1.73	$+6.00$	$+7.92$
8	0.5	$+0.29$	-1.58	$+6.03$	$+7.93$
2	0.25	$+0.25$	-1.67	$+5.96$	$+7.90$
2	0.75	$+0.28$	-1.63	$+6.01$	$+7.91$
2	0.9	$+0.29$	-1.62	$+6.01$	$+7.91$
1	1 (Theorem 1)	$+0.27$	-1.59	$+5.98$	$+7.90$

5.4 Robustness

Backbone seeds. Repeating the TabPFN runs with two more backbone seeds gives per-stream effects that correlate with seed 0 at $r = 0.975$ and 0.973 on real streams and 0.999 on synthetic ones (mean absolute difference 0.56 and 0.64 points); every verdict is unchanged (Appendix F). This concerns the randomness of the backbone only; the variability of the data is covered by the 19 sources and the synthetic seeds.

Hedge hyperparameters. Varying one hyperparameter at a time around the default, $\eta \in \{0.5, 1, 2, 4, 8\}$ with $\gamma = 0.5$ and $\gamma \in \{0.25, 0.5, 0.75, 0.9\}$ with $\eta = 2$ (TabPFN only), the hedge's real-stream mean stays between $+0.24$ and $+0.29$, its worst source between -1.58 and -1.73, and its synthetic gain between $+7.86$ and $+7.93$ (Table 6). The variant covered by Theorem 1 ($\eta = 1$, $\gamma = 1$) behaves the same ($+0.27$, -1.59, $+7.90$). For this variant we also checked the bound of Theorem 1(ii) on every stream and detector: it holds in all 648 cases, the largest case uses 68% of the allowance $N \ln 2$, and in 424 cases the hedge has a lower cumulative log-loss than FIFO.

6 Related work

Drift detection and adaptation. Error-rate detectors with a reset or retrain action are the core of many stream learners (Gama et al., 2004; Baena-García et al., 2006; Bifet & Gavaldà, 2007; Frías-Blanco et al., 2015; Pesaranghader & Viktor, 2016; Raab et al., 2020); surveys discuss their false alarms and delays (Gama et al., 2014; Lu et al., 2019). Ensembles such as adaptive random forests (Gomes et al., 2017) already soften resets by growing a background learner before replacing a member; the hedge is the in-context analogue of that idea with an explicit guarantee.

Tabular foundation models under shift. TabPFN and its successors (Hollmann et al., 2023; 2025; Qu et al., 2025; Ma et al., 2024) learn in context. Drift-Resilient TabPFN (Helli et al., 2024) trains a prior with temporal shifts so that the model handles shift in a single pass over time-indexed data; we instead keep the model fixed and study what the stream controller around it should do.

Prediction with expert advice. The hedge is the aggregating algorithm with two experts (Vovk, 1990; Cesa-Bianchi & Lugosi, 2006) restarted at each alarm; tracking-the-best-expert methods (Herbster & Warmuth, 1998) handle changes without an external detector and are a natural next step.

7 Limitations

The average effect of resets over real sources is not significant, so we do not claim that resets are harmful in general; the claim is about the tail. The real streams are fixed sequences, so the backbone seeds do not vary the data; we rely on 19 sources and synthetic seeds for that. Detectors use default parameters; tuning could change alarm frequency and the tradeoff between gains and losses, which we have not evaluated. The

hedge requires up to twice as many distinct context predictions while the reset and FIFO contexts differ; identical contexts share their cached prediction. Its theorem covers $\eta \leq 1$ without discounting, while the default uses $\eta = 2, \gamma = 0.5$; we show empirically that the choice does not matter here, but the guarantee is for the idealised variant. The trained learners have no probability-based metrics in our study.

8 Conclusion

A FIFO window already makes a tabular in-context learner forget. Adding an error-driven reset on top of it buys little and occasionally costs a great deal. Keeping the FIFO context and hedging against it keeps what the detector is good for and substantially reduces the observed losses. The undiscounted variant has a cumulative log-loss guarantee; the default discounted variant has empirical support here, at the cost of an additional context prediction while the two contexts differ.

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A Proofs

A.1 Proof of Proposition 1

Fix a segment $\tau_j \le t < \tau_{j+1}$. By the definition of RESET, $\mathrm{lo}_t = \max(s_{\tau_j}, \mathrm{lo}_t^F)$ with $s_{\tau_j} = (\tau_j - 1)B$, so $\mathrm{lo}_t \neq \mathrm{lo}_t^F$ exactly when $s_{\tau_j} > \max(0, tB - M)$. This requires $tB - M < (\tau_j - 1)B$, that is $t < \tau_j - 1 + M/B$, which for integer t means $t \le \tau_j + H - 2$ (the condition is necessary, not sufficient: during warm-up both contexts can start at row 0). This proves (i). By Assumption 1 the prediction of batch t depends only on (lo_t, t), so $a_t = a_t^F$ whenever the contexts agree. Summing $a_t - a_t^F$ over each segment, only the batches $\tau_j \le t \le \min(\tau_j + H - 1, \tau_{j+1}) - 1$ contribute, which gives (ii); before τ_1 the two policies coincide. Each Δ_j is a sum of at most $H - 1$ differences of accuracies in $[0, 1]$, which gives (iii). □

A.2 Proof of Theorem 1

Lemma 1 (One batch)**.** *Let $w_F, w_R \ge 0$ with $w_F + w_R = 1$, $p_{k,i} \in (0, 1]$ for $k \in \{F, R\}$ and $i = 1, \dots, B$, $\ell_k = -\frac{1}{B}\sum_i \log p_{k,i}$ and $\eta \in (0, 1]$. Then*

$$-\frac{1}{B}\sum_{i=1}^{B} \log\Big(\sum_k w_k p_{k,i}\Big) \;\le\; -\frac{1}{\eta}\log \sum_k w_k e^{-\eta \ell_k}. \tag{7}$$

Proof. Write $a_i = \sum_k w_k p_{k,i}$. Since $e^{-\eta \ell_k} = \prod_i (p_{k,i}^{\eta})^{1/B}$, the generalised Hölder inequality with B exponents equal to B and weights w_k gives

$$\sum_k w_k \prod_{i=1}^{B} \big(p_{k,i}^{\eta}\big)^{1/B} \le \prod_{i=1}^{B} \Big(\sum_k w_k p_{k,i}^{\eta}\Big)^{1/B}. \tag{8}$$

Since $x \mapsto x^{\eta}$ is concave for $\eta \le 1$, Jensen's inequality gives $\sum_k w_k p_{k,i}^{\eta} \le a_i^{\eta}$. Hence $\sum_k w_k e^{-\eta \ell_k} \le \prod_i a_i^{\eta/B}$; taking $-\frac{1}{\eta} \log$ of both sides proves the claim. □

Proof of Theorem 1. Fix a segment S_j and let $\Lambda_{k,t} = \sum_{u \in S_j, u<t} \ell_{k,u}$ ($\gamma = 1$) and $W_t = \frac{1}{2} \sum_k e^{-\eta \Lambda_{k,t}}$, so $W_{\tau_j} = 1$ and $w_{k,t} = \frac{1}{2} e^{-\eta \Lambda_{k,t}} / W_t$. Applying Lemma 1 with these weights,

$$\ell_{\mathrm{mix},t} \le -\frac{1}{\eta} \log \sum_k w_{k,t} e^{-\eta \ell_{k,t}} = -\frac{1}{\eta} \log \frac{W_{t+1}}{W_t}. \tag{9}$$

Summing over $t \in S_j$ telescopes to $-\frac{1}{\eta} \log W_{\mathrm{end}}$, and $W_{\mathrm{end}} \ge \frac{1}{2} e^{-\eta \Lambda_{k,\mathrm{end}}}$ for each k, which gives (i). For (ii), on S_0 the experts coincide, so the mixture equals FIFO with no additive cost; on each of the N later segments (i) bounds the mixture by the better expert, and in particular by F, plus $\ln 2/\eta$. Summing over segments gives both bounds. □

B Data

Table 7: The 19 real sources: number of streams (segments), total rows, features and classes.

Source	Streams	Rows	Features	Classes
airlines	5	500,000	7	2
bank	1	41,188	19	2
chest	1	100,000	3	7
covertype	5	550,000	54	7
eeg	1	14,980	14	2
elec2	1	45,312	8	2
gas	1	13,910	128	6
home	1	100,000	10	3
insects	9	787,311	33	6
kdd	1	100,000	41	5
news	1	39,644	58	2
occupancy	1	20,560	5	2
phishing	1	11,055	46	2
poker	5	500,000	10	10
rialto	1	82,250	27	10
room	1	10,129	16	4
spam	1	6,213	499	2
wall	1	5,456	24	4
weather	1	18,159	8	2

C Reproducibility details

Real streams. All streams are stored as feature matrix, integer label and (unknown) concept index; rows keep the order of the source, labels are numbered in sorted order, categorical columns are integer-coded, and time columns are used only for ordering. Before any model was run, each held-out and added stream was checked with a sliding random forest against the majority-class rate; for the ten added sources a majority rate of at least 0.98 was a pre-set exclusion rule, which excluded none.

- *Electricity* and the seven *INSECTS* variants (Souza et al., 2020): the `river` 0.26.1 datasets `Elec2` and `Insects`, INSECTS features parsed as numbers, used in full except that covertype and the INSECTS abrupt and gradual imbalanced streams are cut to their first 150 000 rows.

- *airlines, phishing, poker, rialto, spam, weather* (prefix `h2_`): every real stream of the `ogozuacik/concept-drift-datasets-scikit-multiflow` collection with at least 5 000 rows and at most 10 classes not used before, in file order, cut to the first 100 000 rows.

- Later *segments* of the long streams, rows never used before: airlines and poker rows $[100\text{k}, 200\text{k})$, $[200\text{k}, 300\text{k})$, $[300\text{k}, 400\text{k})$, $[400\text{k}, 500\text{k})$; covertype rows $[150\text{k}, 250\text{k})$, $[250\text{k}, 350\text{k})$, $[350\text{k}, 450\text{k})$, $[450\text{k}, 550\text{k})$; INSECTS abrupt-imbalanced rows $[150\text{k}, 250\text{k})$, $[250\text{k}, 350\text{k})$.

- The ten added sources, in time order and cut to the first 100 000 rows: Gas Sensor Array Drift (UCI 224, batches 1–10), Occupancy Detection (UCI 357), Room Occupancy Estimation (UCI 864), Bank Marketing (UCI 222, file order, `duration` removed), KDD Cup 1999 10% (via `scikit-learn`, labels grouped into normal and the four attack families), EEG Eye State (UCI 264), Online News Popularity (UCI 332, oldest article first, label shares ≥ 1400), Gas Sensors for Home Activity Monitoring (UCI 362, recordings in calendar order, all rows kept; rows outside the stimulus interval are labelled background), Wall-Following Robot Navigation (UCI 194, 24 sensors) and the Single Chest-Mounted Accelerometer data (UCI 287, subjects 1–15 concatenated, unlabelled rows removed).

A source's value is the mean over its streams (Table 7).

Synthetic streams. Recurring `river` streams visit K fixed concept generators in blocks of 2 000 rows and cycle through them three times; a recurring concept continues its own random stream, so it has the same distribution but new rows. Test seeds $s = 10, \ldots, 14$; each concept generator is seeded with $10s$ plus its variant or function index (for Agrawal the functions 0, 2, 4, 6; for RBF and hyperplane the concept index). SEA (variants 0–2, $K = 3$, 18 000 rows), STAGGER (functions 0–2, $K = 3$, 18 000), Agrawal (functions 0, 2, 4, 6, $K = 4$, 24 000), sine (functions 0–3, $K = 4$, 24 000) change only the labelling function; RBF (20 centroids, 4 classes, 10 features, $K = 3$, 18 000) and hyperplane (10 features, 5% noise, $K = 3$, 18 000) change both inputs and labels. The twelve grid streams have three concepts that share an RBF mixture of 5, 30 or 100 centroids in 10 dimensions with 4 classes and differ only in the label of each centroid; blocks of 500 or 2 000 rows, three cycles, seeds 10 and 11.

Models. TabPFN v2 (`tabpfn` 9.0.0, default v2 classifier checkpoint, `n_estimators=4`, `ignore_pretraining_limits`), TabICL (`tabicl` 2.2.0, `n_estimators=4`) and TabDPT (`tabdpt` 1.3.1, four ensemble members), all with PyTorch 2.14 on one GPU; backbone seed 0 unless stated, passed as `random_state` to TabPFN and TabICL and as the ensemble seed of `ensemble_predict_proba` to TabDPT. Trained learners and detectors use `river` 0.26.1.

Prediction details. Classes absent from a context receive probability 0; a context with a single class predicts that class with probability 1. These cases violate Assumption 2; log-losses are computed with probabilities clipped at 10^{-6}. Predictions are cached in half precision per (context start, batch) and the mixture of the hedge is renormalised before scoring. Batch 0 is not scored, and a final incomplete batch is dropped.

Detectors. One detector object per policy run, default parameters (KSWIN with `seed=0`). After the labels of a batch arrive, it receives the per-row 0/1 errors of the policy's own final prediction (for HEDGE, the mixture); feeding stops at the first alarm in a batch, and `river` re-initialises the detector's statistics at its next update. On an alarm, HEDGE sets the discounted loss of the reset expert to that of FIFO, so both weights become 1/2.

D Metrics beyond accuracy

The result files of the main runs store per-batch accuracy and log-loss. Because every policy is a deterministic function of the cached TFM predictions (Assumption 1), we replay each policy from the cache with the same code path, check that the replayed per-batch accuracies and reset positions equal the stored ones exactly, and compute the metrics over all scored rows. In 17 streams, 721 cache entries in total had been lost to concurrent writes of the same cache file (before writes were made to merge); they were recomputed with the TFM, and the replay check confirms that the recomputed predictions reproduce the stored per-batch accuracies and reset positions exactly.

E Trained learners

Table 8 gives the absolute accuracy of the never-reset learners and the per-source effect of a full reset for the trained learners and TabPFN.

Table 8: Left: accuracy (%) of the never-reset learner, source-weighted. Right: effect of a full reset per real source, averaged over the 8 detectors (points).

Learner	Real	Synthetic
Naive Bayes	61.9	64.8
Hoeffding tree	72.4	67.6
TabPFN	87.8	83.9

Source	NB	HT	TabPFN
wall	−9.9	−9.2	−18.8
room	−6.2	−0.8	−2.5
occupancy	−3.7	−3.5	−1.5
covertype	−3.3	−17.5	−23.1
phishing	−0.6	−0.7	0.0
weather	0.0	−3.1	−1.0
news	+0.1	−3.2	−0.5
airlines	+0.5	−1.7	−0.5
elec2	+1.7	−3.1	−1.4
rialto	+3.6	+4.0	+0.8
kdd	+4.2	+2.4	−0.2
poker	+5.8	+3.8	−8.3
insects	+8.5	+4.0	+0.5
spam	+9.2	−1.8	−0.8
gas	+9.8	+8.9	−6.4
bank	+22.1	+0.3	0.0
chest	+23.4	+14.5	+1.9
home	+34.1	+12.5	+1.4
eeg	+39.2	+28.5	+6.3

F Pre-registered hypotheses and outcomes

Table 9 lists, stage by stage, the criteria fixed before the stage was run and the outcomes, including the ones that failed.

Table 9: Hypotheses and pass criteria fixed before each stage, and their outcomes. The synthetic streams of the TabDPT and $M = 2000$ stages were run after their real streams, with the same criteria. The tail tests T1–T2 were added after the first TabPFN run (they were suggested by it) and were then tested on the later stages.

Stage	Criterion	Outcome
TabPFN, 29 MICE streams	H1: full reset below FIFO on real for $\geq 6/8$ detectors; H2: above on synthetic for $\geq 6/8$; H3: $> 50\%$ of single resets lose on real and win on synthetic; H4: HEDGE ≥ -0.2 on real and keeps $\geq$ half of the synthetic gain	all hold
10 new sources (two batches of 5)	same direction on the added sources	first batch: yes, but **H4 fails there** (HEDGE -0.4 to -0.9); **second batch: opposite** (resets help on 3 of 5); over 19 sources no detector significant after Holm
TabICL	H1'–H4' over 19 sources; T1: HEDGE beats RESET on the worst source by ≥ 10; T2: HEDGE keeps $\geq$ half of the best-source gain	all hold
TabPFN seeds 1, 2	S1: per-stream effects correlate with seed 0 at $r \geq 0.9$; S2: signs kept for $\geq 7/8$ detectors, T1–T2 hold	all hold
Hedge sensitivity	HS1–HS3 (HEDGE criteria of H4 and T1) for all 7 variants	7/7 hold (expected to fail for $\eta = 0.5$; did not)
Trained learners	TR1: learner's full-reset mean ≥ 2 points above TabPFN's; TR2: worst source ≥ 10 points better; TR3: $< 70\%$ harmful resets	NB: all hold; HT: TR1, TR3 hold, **TR2 fails** (8.8)
$M = 500$, $M = 2000$	H1'–H4', T1, T2; prediction: both gain and loss grow with M	all hold; prediction **only half** confirmed (loss flat from 1000 to 2000)
TabDPT	D1: T1, T2 hold; D2: full-reset mean negative and $> 70\%$ harmful resets	all hold